



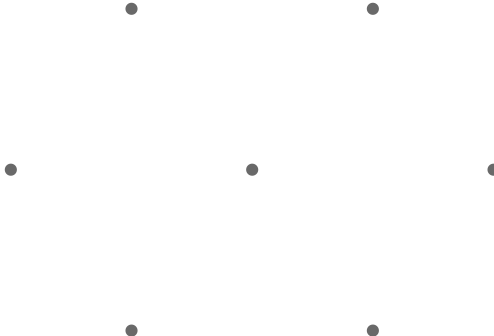
Technische  
Universität  
Braunschweig



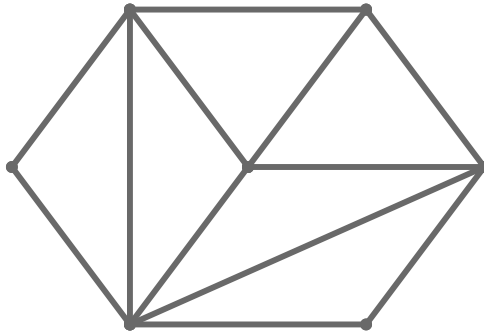
# Algorithmische Methoden zu gradbeschränkten Triangulierungen

Fabian Alich, 3. September 2025

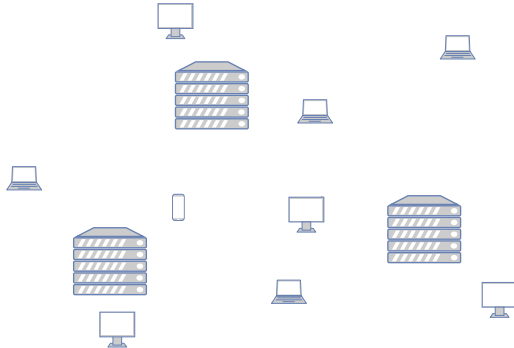
# Triangulierung



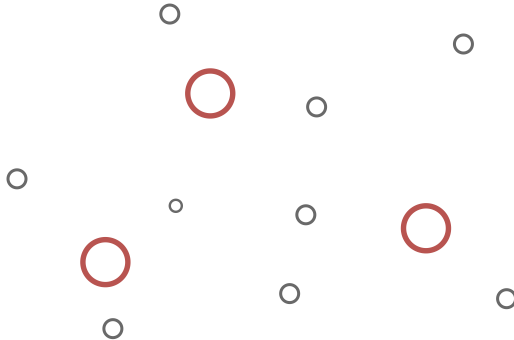
# Triangulierung



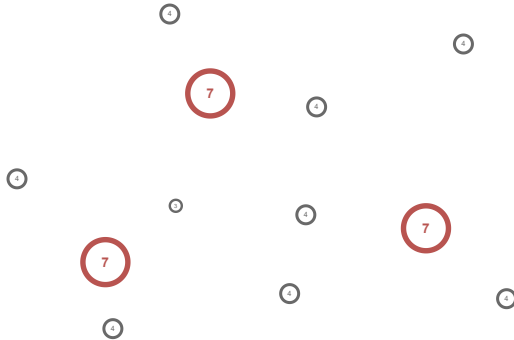
# Praktische Relevanz



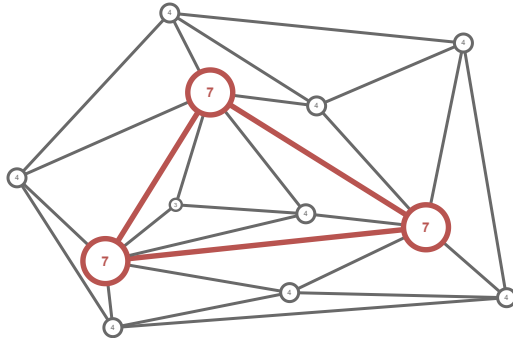
# Praktische Relevanz



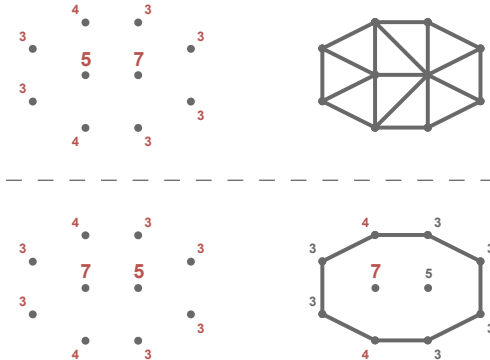
# Praktische Relevanz



# Praktische Relevanz



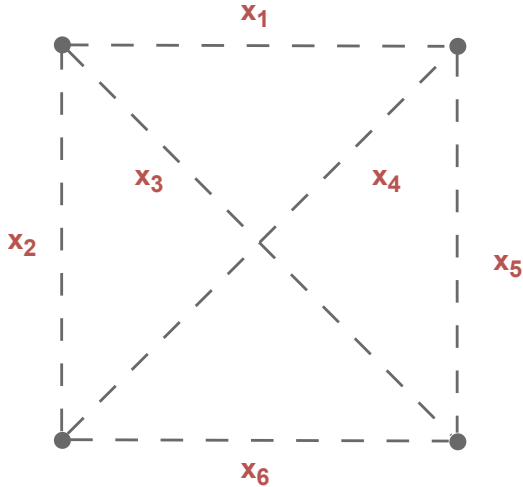
# Das Problem



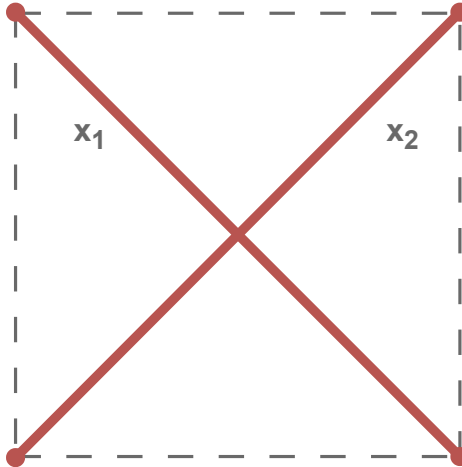
# Teil I

## Lösungsansätze

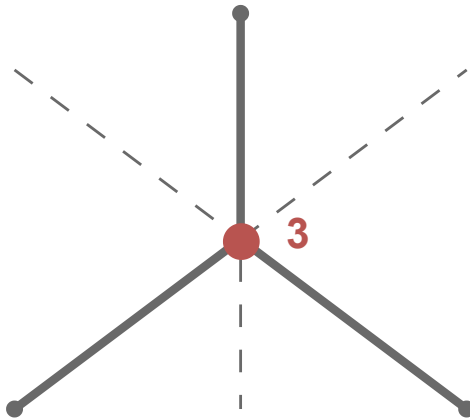
# Kantenansatz



# Überschneidungsbedingung



# Gradbeschränkung



# Maximale Kanten

FA: Formal raus?

$$\text{Anzahl Kanten} \cdot 2 \stackrel{!}{=} \sum \text{Sollgrad}$$

# Gradbeschränkung

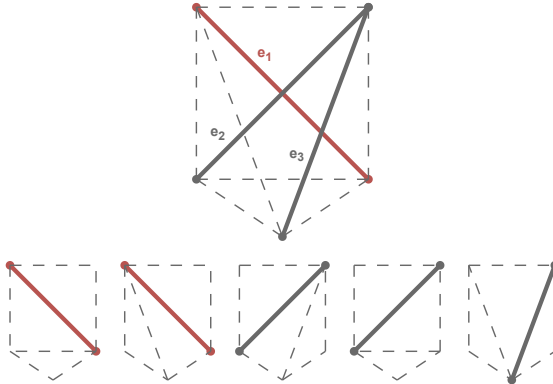
FA: Durchschnitt

# Gradbeschränkung

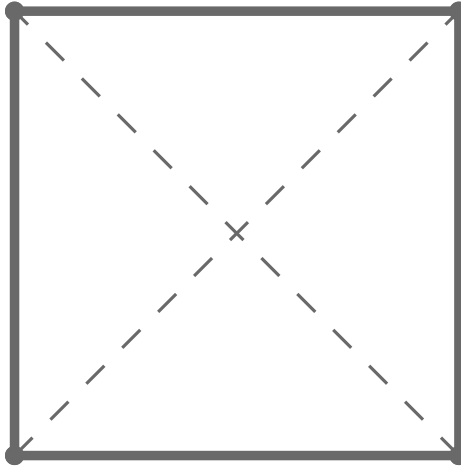
FA: Max min

# alternative Kantenbedingung

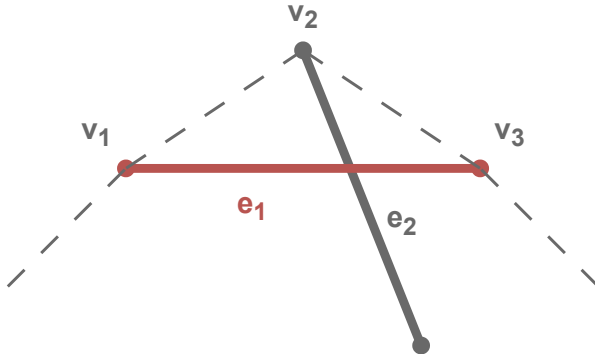
FA: sollte ich hier eine Referenz auf euer Paper machen?



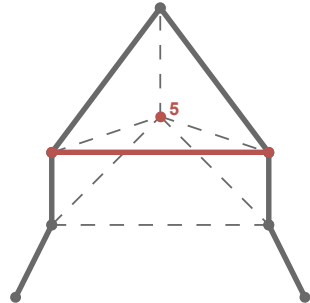
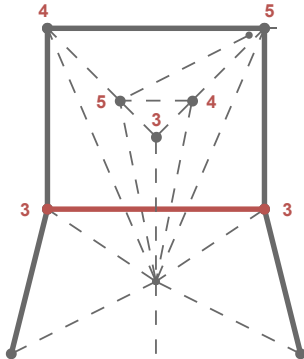
# Hülle Fixieren



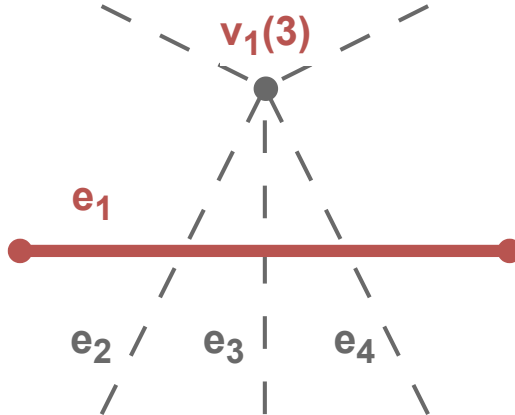
# Kanten Fixieren / Ausschließen



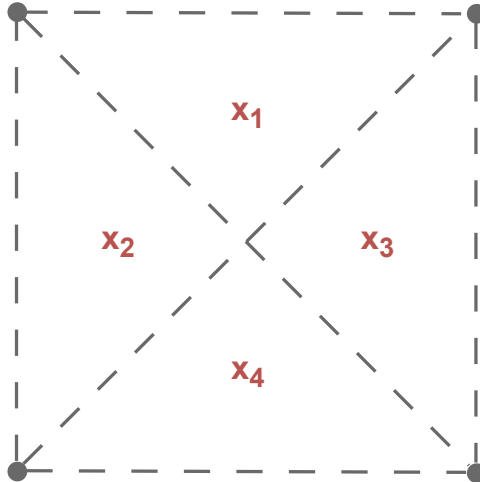
# Kanten Ausschließen



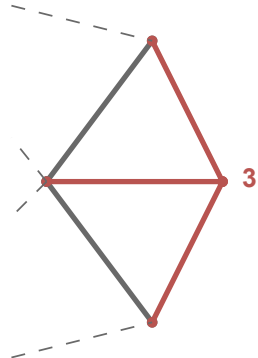
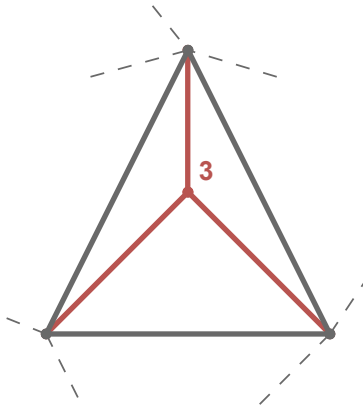
# Kanten Ausschließen



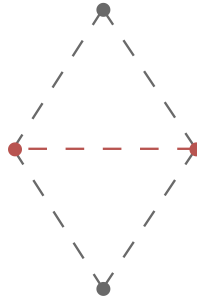
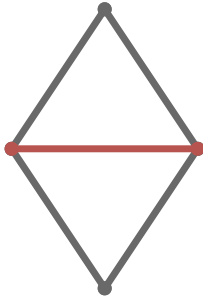
# Dreiecksansatz



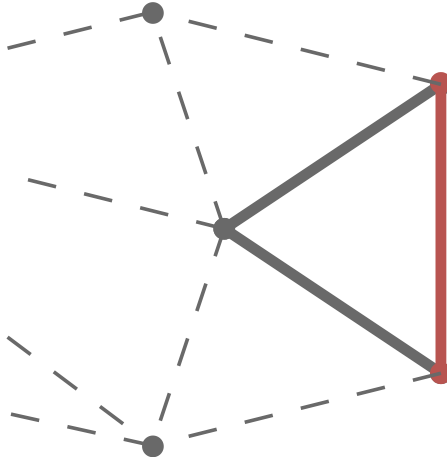
# Gradbeschränkung



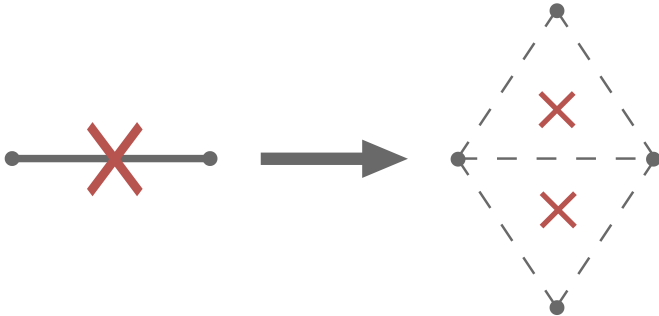
# Überschneidungsbedingung



# Überschneidungsbedingung



# Dreiecks Ausschließen



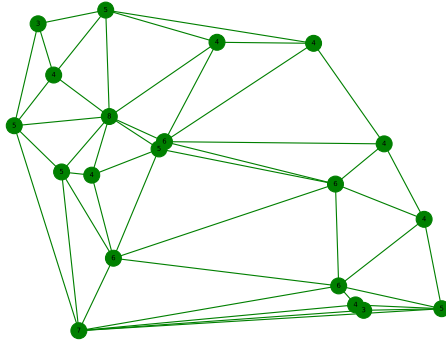
# Subset

FA: ich habe jetzt subset in den Diagrammen drin, ist aber schlecht. Soll ich das dann hier erklären oder wie mache ich das am besten? Einfach nur auf die Arbeit verweisen?

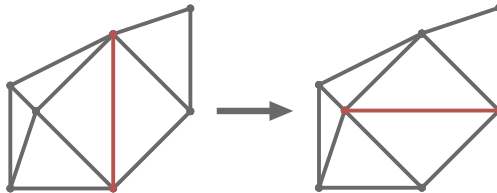
## Teil II

# Instanzgenerierung

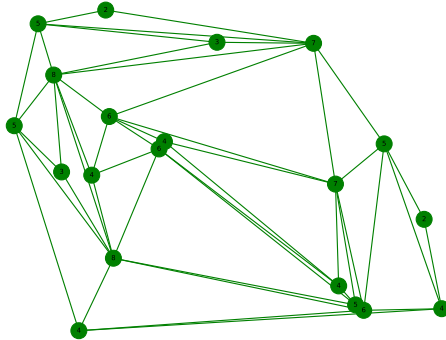
# Delaunay



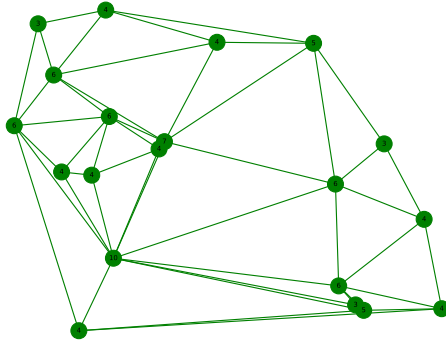
# Flip



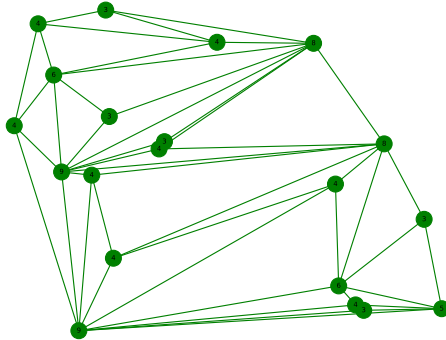
# Delaunay-Flips



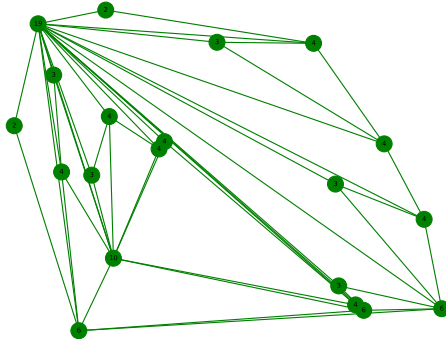
# Greedy



# Random



# Iterativ



# Move

FA: Bild erstellen

# Change

FA: Bild erstellen

# Teil III

## Auswertung

# Solver



**Gurobi**

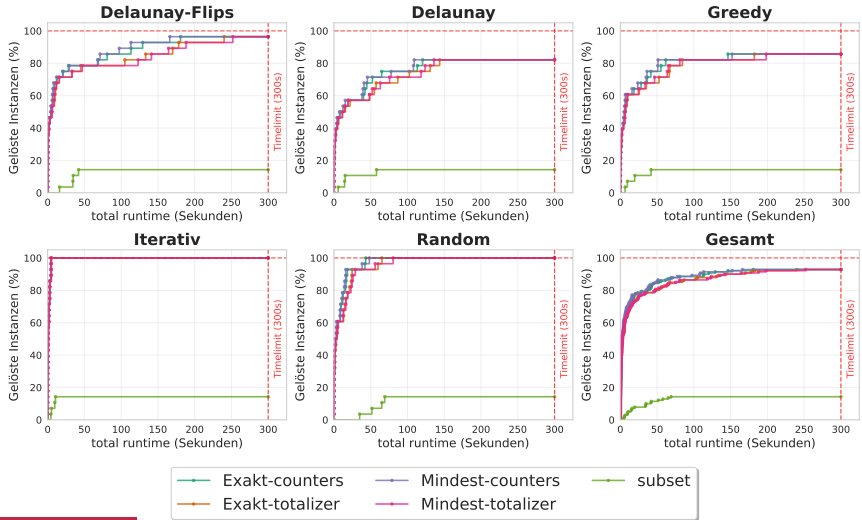


**OR-Tools**

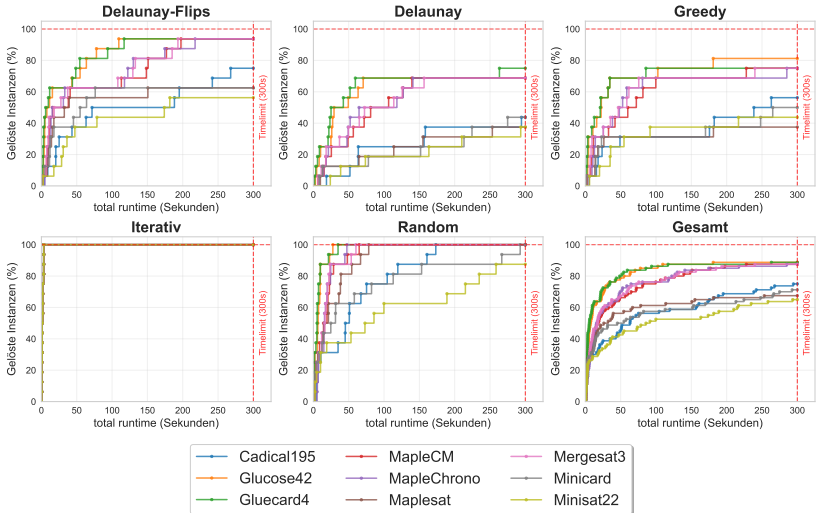


**PySAT**

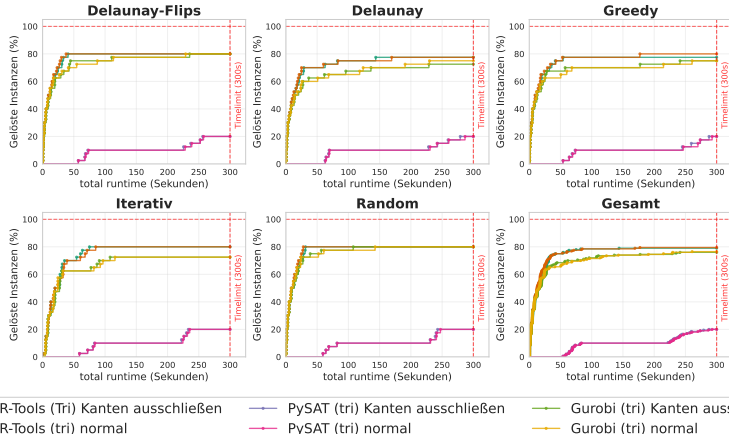
# Grad Codierung



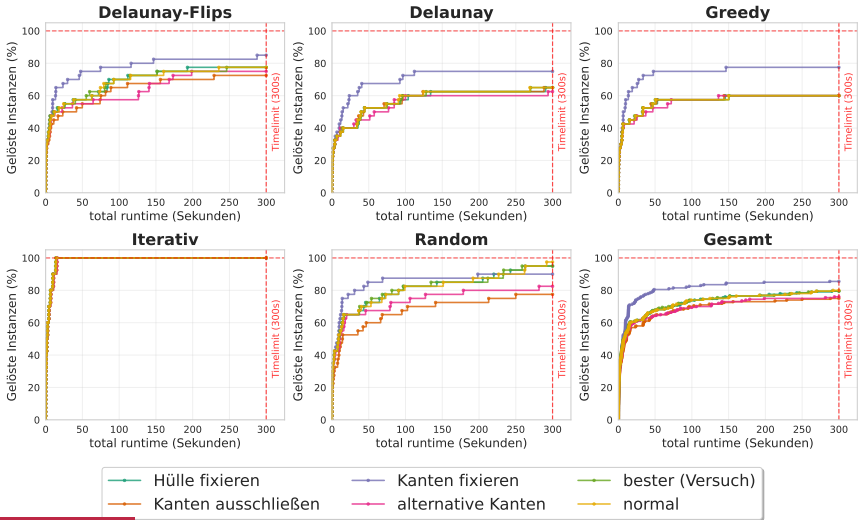
# PySAT Solver



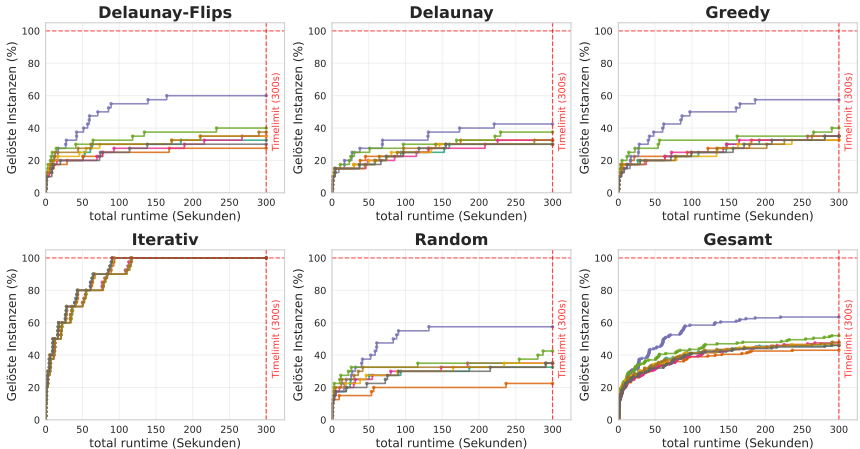
# Dreiecke Bedingungen



# PySAT Bedingungen

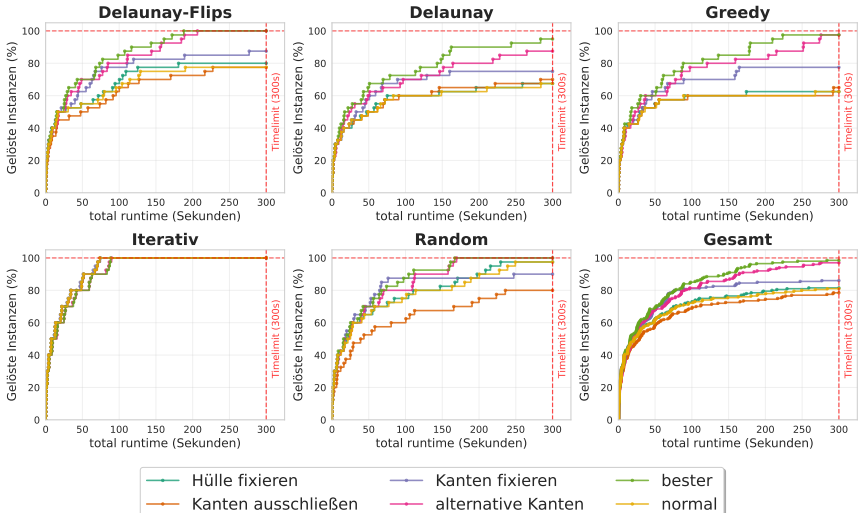


# Gurobi Bedingungen

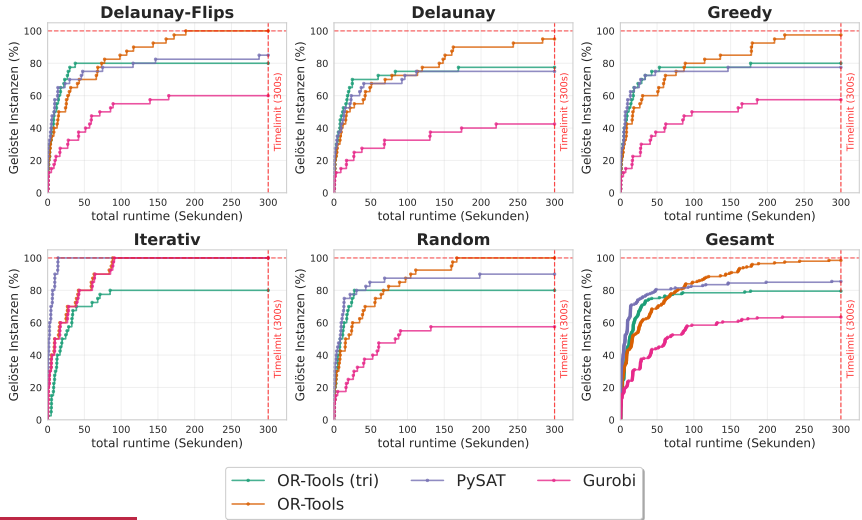


- Hülle fixieren
- Kanten ausschließen
- Kanten fixieren
- alternative Kanten
- bester (Versuch 1)
- bester (Versuch 2)
- bester (Versuch 3)
- normal

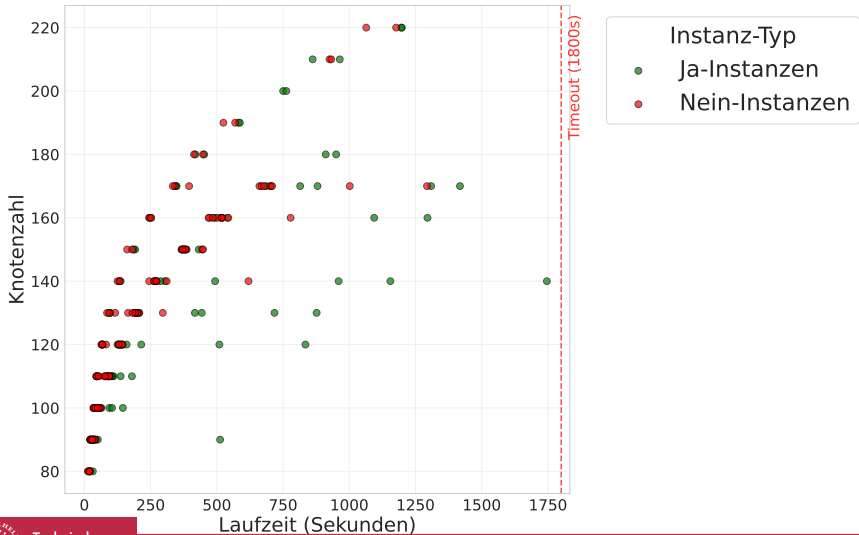
# OR-Tools Bedingungen



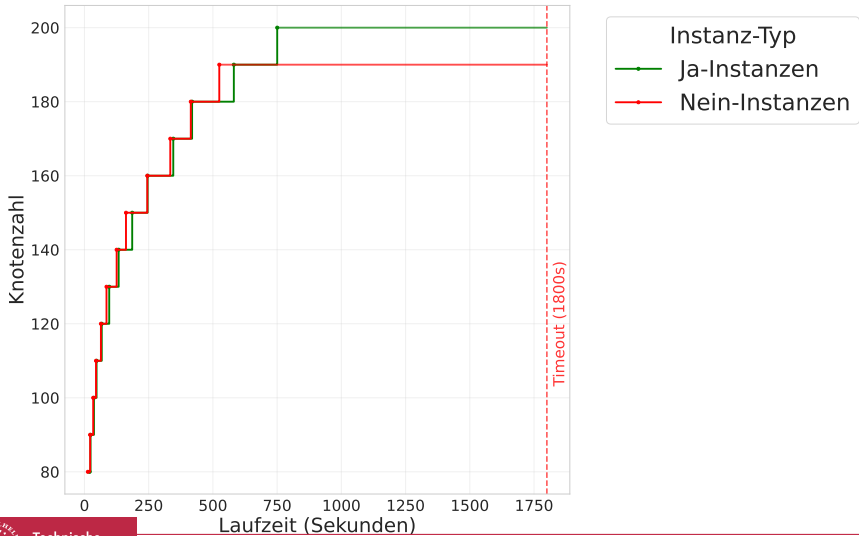
# Solver Vergleich



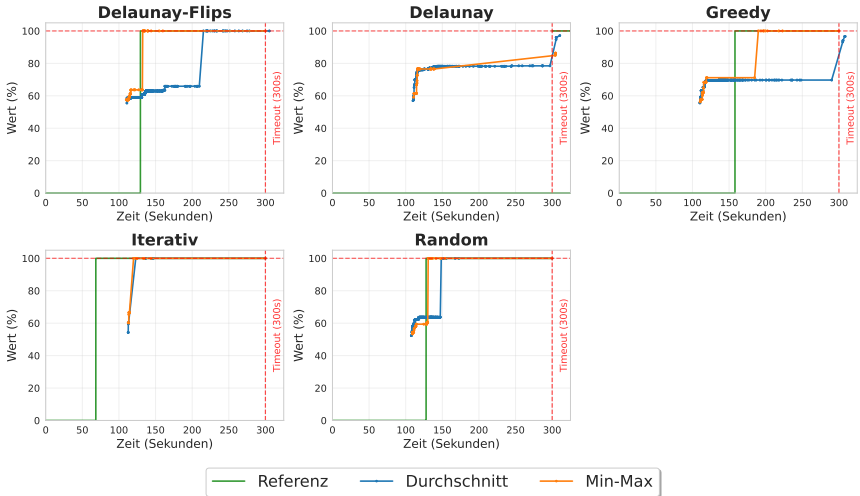
# Größerer Timeout



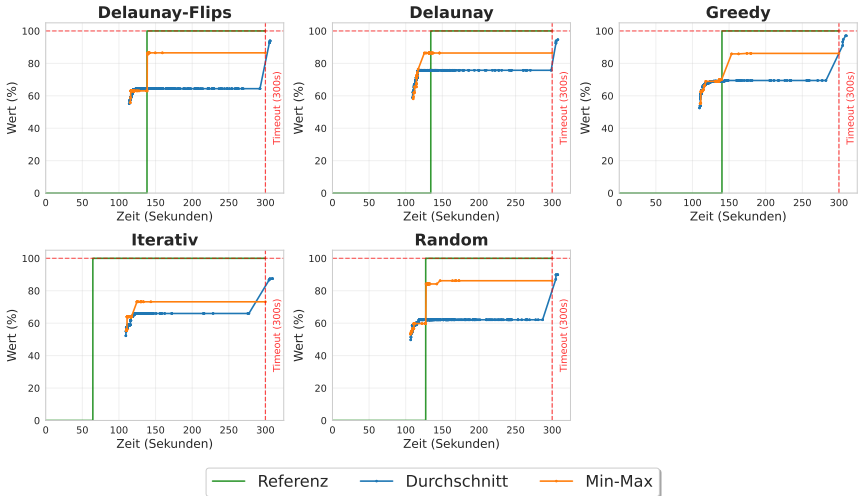
# Ja Nein getrennt



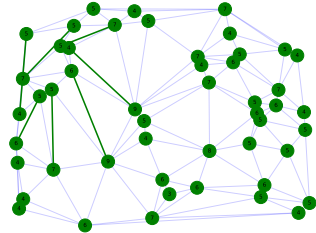
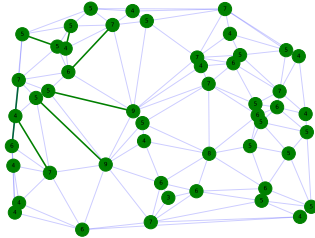
# Ja Instanz



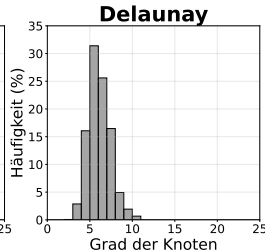
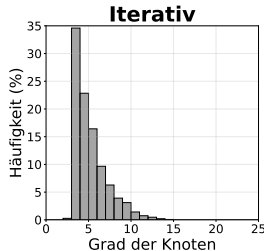
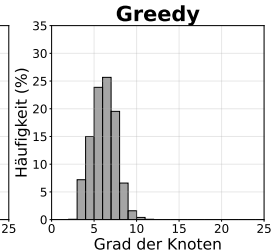
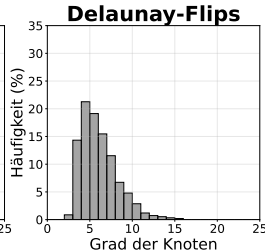
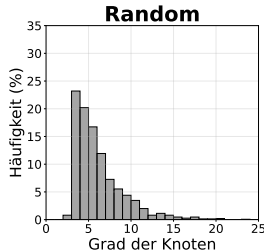
# Nein Instanz



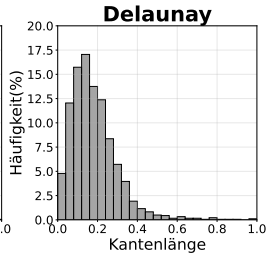
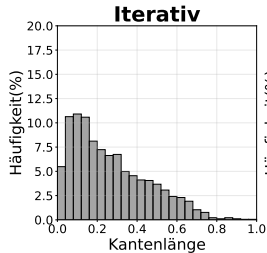
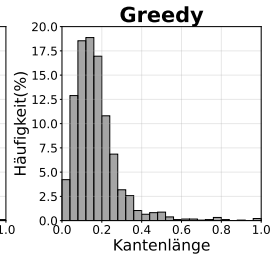
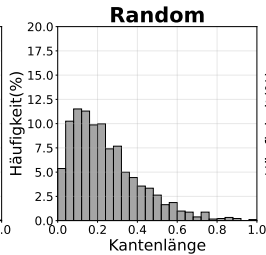
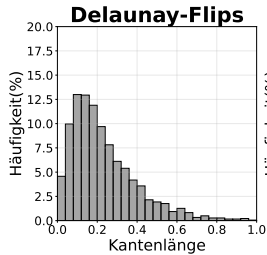
# Anzahl Lösungen



# Gradverteilung



# Kantenlängen

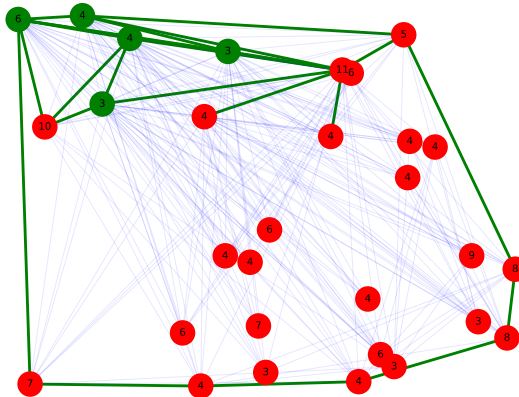


## CaDiCaL Simplified Satisfiability Solver

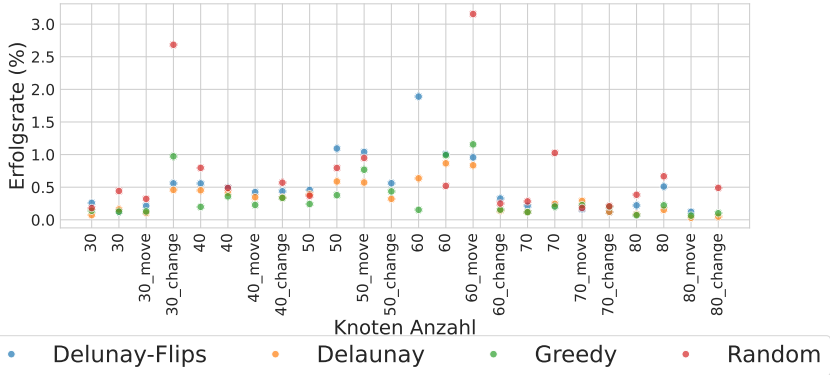
# Teilinstanz

FA: Isolierte Kante Teilinstanz einfügen

## Teilinstanz



# Propagation



## Teil IV

# Fazit und Ausblick

# Fazit

## OR-Tools Hülle fixieren Kanten Fixieren alternative Kantenbedingung



# Fazit

FA: Bild von Instanze mit 220 Knoten

# Fazit

FA: Bild von Optimierungsproblem mit 97 Prozent

# Fazit

FA: Bild wo eine Propagation möglich war

# Ausblick

- Skalierbarkeit
- Arbeitsspeicherverbrauch
- Dreiecksansatz Schwierigkeit Instanzen
- Optimierungen Kombinationen
- Propagation erweitern
- normalverteilte Triangulierung
- Problem abwandeln
  - Kanten vorgeben
  - Polygone